

FUTURE INTERNS

Cybersecurity Internship — Task 2

PHISHING EMAIL DETECTION & AWARENESS REPORT

Prepared by: NISHANTH N NAIK

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1.INTRODUCTION

Phishing is one of the most dangerous and widespread cyber threats targeting individuals and organizations today. Attackers send fake emails pretending to be trusted companies or officials to trick victims into clicking malicious links, sharing passwords, or transferring money. This report presents a detailed analysis of three real phishing email samples collected from a public security research repository. Each email was examined using professional tools including Google Header Analyzer, MXToolbox, and VirusTotal to identify attack indicators and confirm malicious intent. The findings are documented along with prevention guidelines to help employees and organizations defend against phishing attacks.

2.METHODOLOGY

1. Sample Collection: Real phishing emails were collected from public repository github.com/rf-peixoto/phishing_pot
2. Header Extraction: The raw email header was extracted from each .eml file by opening it in Raw view on GitHub.
3. Header Analysis: Each header was analyzed using:
 - Google Admin Toolbox Message Header Analyzer
 - MXToolbox Email Header Analyzer

4. Authentication Checks: SPF, DKIM, and DMARC results were recorded for each email to detect spoofing.
5. URL Scanning: Suspicious links found in email bodies were safely scanned using VirusTotal.
6. Risk Classification: Each email was rated as Safe / Suspicious / Phishing based on findings.

3.Email Analysis

3.1 Email Sample 1 — Inheritance Scam (Advance Fee Fraud)

Field	Details
File	sample-1720.eml
From (Claimed)	"Mr.UKA Ezekiel" dianaf65@icloud.com
To	Recipients dianaf65@icloud.com
Subject	Attention
Date	Friday, 27 October 2023
Real Sending Server	mail.decipherhealth.in
Real IP	103.2.233.28

SPF	Softfail ✖
DKIM	None ✖
DMARC	Fail ✖
Delivery Delay	68 minutes (suspicious)
Risk	● PHISHING

Subject: Attention
To: Recipients <dianaf65@icloud.com>
From: "Mr.UKA Ezekiel" <dianaf65@icloud.com>
Date: Fri, 27 Oct 2023 08:40:18 -0700
Reply-To: ftom1346@gmail.com
Message-Id: <20231027154023.CCE7FB87BB8@mail.decipherhealth.in>

Figure 1: Email header showing From, Subject, To, Reply-To fields

The screenshot shows the Google Admin Toolbox Messageheader interface. The top section displays the email header details, including the MessageId, Created at, From, To, Subject, and SPF, DKIM, and DMARC status. The bottom section shows a table of delivery details, including the delay, from, to, protocol, and time received.

#	Delay	From *	To *	Protocol	Time received
0	5 sec	unknown	→ mail.decipherhealth.in	ESMTP	10/27/2023, 9:10:23 PM GMT+5:30
1	3 mins	mail.decipherhealth.in	→ localhost	ESMTP	10/27/2023, 9:13:42 PM GMT+5:30
2		localhost	→ mail.decipherhealth.in	ESMTP	10/27/2023, 9:13:42 PM GMT+5:30

Figure 2: Google Header Analyzer showing SPF softfail, DMARC fail, and 68 minute delivery delay

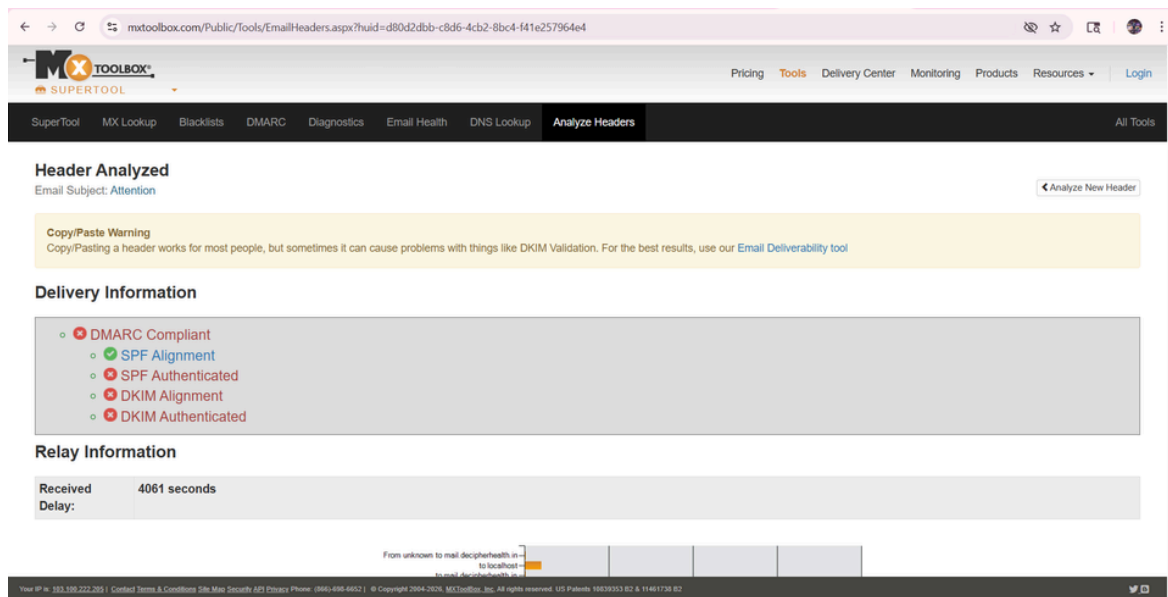


Figure 3: MXToolbox confirming DMARC non-compliant, SPF and DKIM authentication failures

Indicators Found:

- Email spoofing — From and To have the same iCloud address
- Mismatched Reply-To — replies go to ftom1346@gmail.com
- Fake sending server — sent from decipherhealth.in not iCloud
- SPF Softfail — unauthorized server used
- DKIM None — email has no digital signature
- DMARC Fail — overall authentication failed
- 68 minute delivery delay — suspicious routing
- Social engineering — fake \$12.6 million inheritance story
- Personal data harvesting — asks for name, address, phone
- Grammar errors — "You are Advise to Re-confirm"

Risk Classification:PHISHING

3.2 Email Sample 2 — MercadoPago Payment Phishing

Field	Details
File	sample-172.eml
From (Claimed)	Mercado Pago naoresponder@mercadopago.com.br
To	phishing@pot
Subject	Scheduled transfer of R\$980 by Rodrigo F P
Date	Sunday, 18 December 2022
Real Sending Server	soldrun.com
Real IP	139.177.198.148
SPF	Fail ✖
DKIM	Fail ✖
DMARC	Fail ✖
Malicious URL	ii-dnuqnaqtnq-rj.a.run.app
VirusTotal Result	7/95 vendors flagged MALICIOUS
Risk	● PHISHING

35 Content-Type: text/html; charset=utf-8
36 From: Mercado Pago <naoresponder@mercadopago.com.br>
37 To: phishing@pot
38 Subject: =?UTF-8?Q?Transfer=C3=A4ncia_agendada_para_Segunda?=
39 =?UTF-8?Q?-Feira=2C_19_de_Dezembro=2C_=C3=A0s?= 15:13 hs por Rodrigo F P.
40 Message-ID: <b3a28f35-4255-96f5-8a5b-f93438107c69@mercadopago.com.br>
41 Content-Transfer-Encoding: quoted-printable
42 Date: Sun, 18 Dec 2022 18:14:33 +0000

Figure 4: Email details showing From, Subject, Date fields

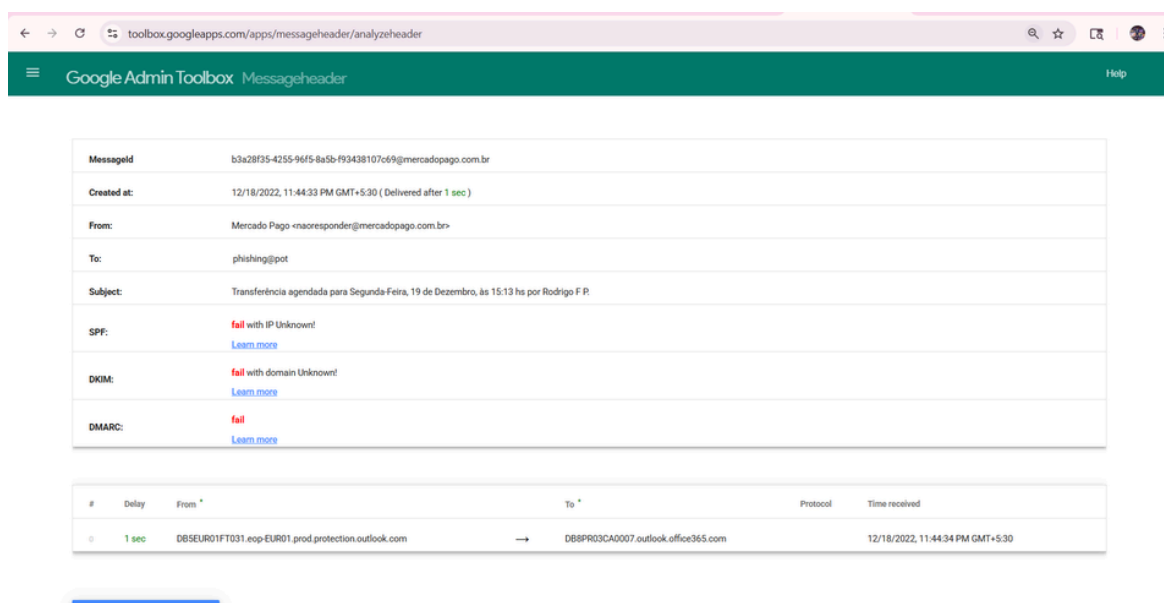


Figure 5: Google Header Analyzer — SPF fail, DKIM fail, DMARC fail

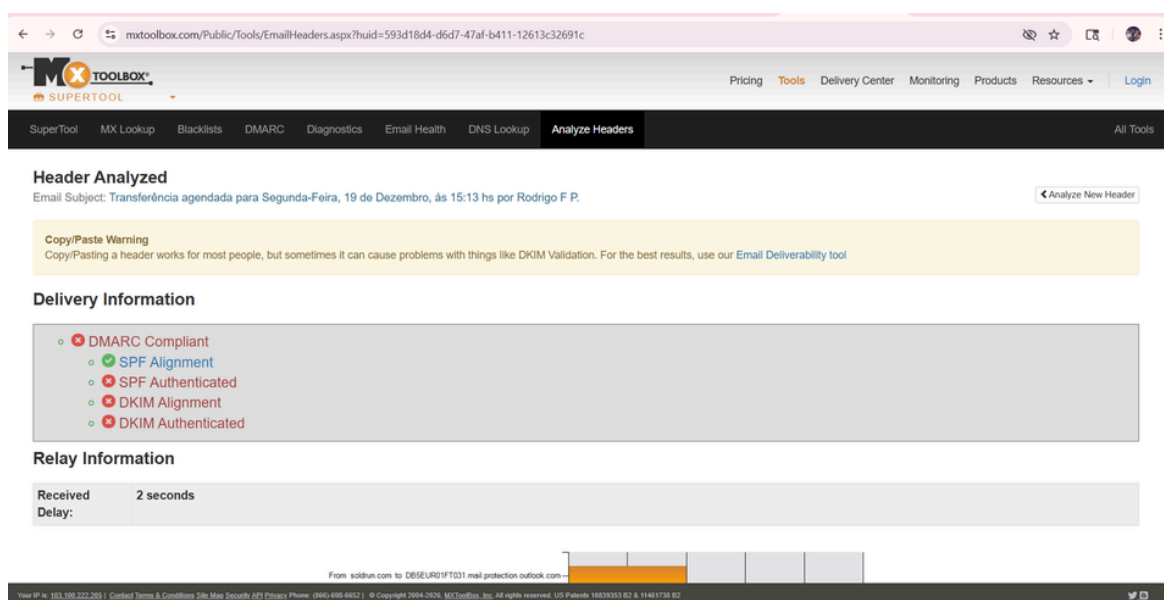
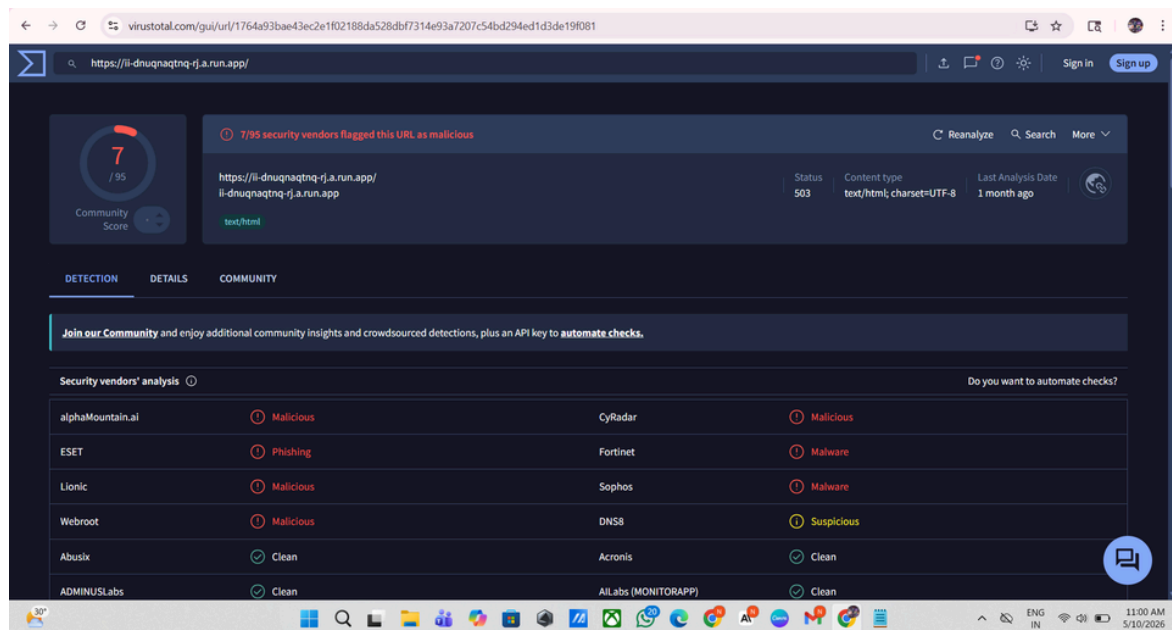


Figure 6: MXToolbox — DMARC non-compliant, SPF and DKIM authentication failed



VirusTotal scan — 7 out of 95 vendors flagged URL as malicious (Phishing/Malware)

Indicators Found:

- Domain spoofing — sent from soldrun.com, claims mercadopago.com.br
- SPF Fail — unauthorized server confirmed
- DKIM Fail — email signature invalid
- DMARC Fail — domain authentication failed
- Urgency tactic — claims R\$980 transfer happening NOW -
- Malicious redirect — all buttons link to ii-dnuqnaqtnq-rj.a.run.app
- VirusTotal confirmed — 7/95 vendors flagged as malicious
- Hidden tracking pixel — 1x1 invisible image to spy on victims
- Spam filter evasion — hidden random words in white text

Risk Classification: PHISHING

3.3 Email Sample 3 — Kohl's Fake Prize Phishing

Field	Details
File	sample-1721.eml
From (Claimed)	KOHL's r5mk7@ri69aezzhy.com
To	phishing@pot
Subject	Confirmation Needed Act Now !
Date	Sunday, 29 October 2023
Real Sending Server	gybatedtioneer.net
Real IP	88.151.12.68
SPF	None ✖
DKIM	None ✖
DMARC	None / No Record Found ✖
Malicious UR-1	t.co/k2FXJOglkU (1/91 flagged)
Malicious URL-2	dtherhproblem.us (1/96 flagged — Phishing)
Risk	● PHISHING

From: KOHL's ,_<r5mk7@ri69aezzhy.com>
Subject: Confirmation Needed Act Now !
To: phishing@pot
Content-Length: 22448528
Content-Length: 49413
Date: Sun, 29 Oct 2023 06:17:27 +0000

Figure 8: Email details — fake Kohl's sender domain ri69aezzhy.com

The screenshot shows the Google Admin Toolbox Messageheader interface. The top section displays the email's metadata: MessageId (baa7b6f4-bdad-4596-a357-1a8493934df0@V11EUR04FT053.eop-eur04.prod.protection.outlook.com), Created at (10/29/2023, 11:47:27 AM GMT+5:30), From (KOHL's ,_<r5mk7@ri69aezzhy.com>), To (phishing@pot), Subject (Confirmation Needed Act Now!), SPF (none), DKIM (none), and DMARC (none). Below this, a table shows the delivery path with two hops. The first hop is from V11EUR04FT053.eop-eur04.prod.protection.outlook.com to AS9PR06CA0498.outlook.office365.com. The second hop is from AS9PR06CA0498.eurprd06.prod.outlook.com to DM4P223MB0518.NAMP223.PROD.OUTLOOK.COM.

#	Delay	From	To	Protocol	Time received
0		V11EUR04FT053.eop-eur04.prod.protection.outlook.com	AS9PR06CA0498.outlook.office365.com		10/29/2023, 11:47:27 AM GMT+5:30
1	2 sec	AS9PR06CA0498.eurprd06.prod.outlook.com	DM4P223MB0518.NAMP223.PROD.OUTLOOK.COM		10/29/2023, 11:47:29 AM GMT+5:30

Figure 9: Google Header Analyzer — SPF none, DKIM none, DMARC none

The screenshot shows the MXToolbox 'Analyze Headers' results for the email subject 'Confirmation Needed Act Now !'. A 'Copy/Paste Warning' is displayed. Under 'Delivery Information', it shows 'DMARC Compliant (No DMARC Record Found)' with sub-items for SPF Alignment, SPF Authenticated, DKIM Alignment, and DKIM Authenticated, all marked with red 'X' icons. Under 'Relay Information', it shows 'Received' as '3 seconds' and 'Delay' as '3 seconds'. At the bottom, a visual representation of the email's path is shown, starting from 'gubatedioneer.net' to 'V11EUR04FT053.mail.protection.outlook.com'.

Header Analyzed
Email Subject: Confirmation Needed Act Now !

Copy/Paste Warning
Copy/Pasting a header works for most people, but sometimes it can cause problems with things like DKIM Validation. For the best results, use our [Email Deliverability tool](#)

Delivery Information

- DMARC Compliant (No DMARC Record Found)
 - SPF Alignment
 - SPF Authenticated
 - DKIM Alignment
 - DKIM Authenticated

Relay Information

Received: 3 seconds
Delay: 3 seconds

Figure 10: MXToolbox — No DMARC record found, SPF and DKIM completely absent

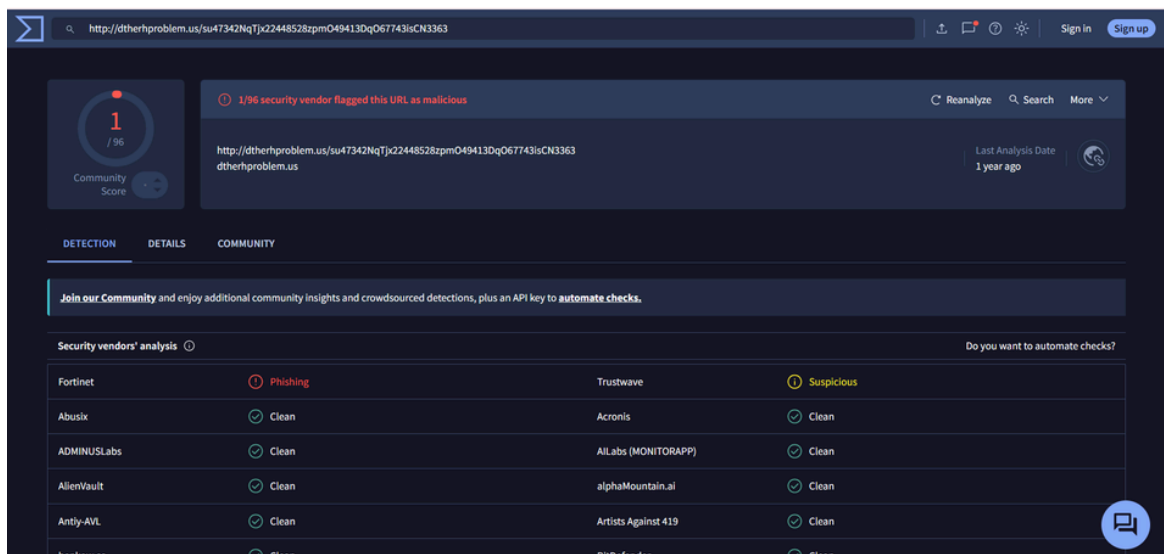


Figure 11: VirusTotal scan of t.co link — flagged as Phishing

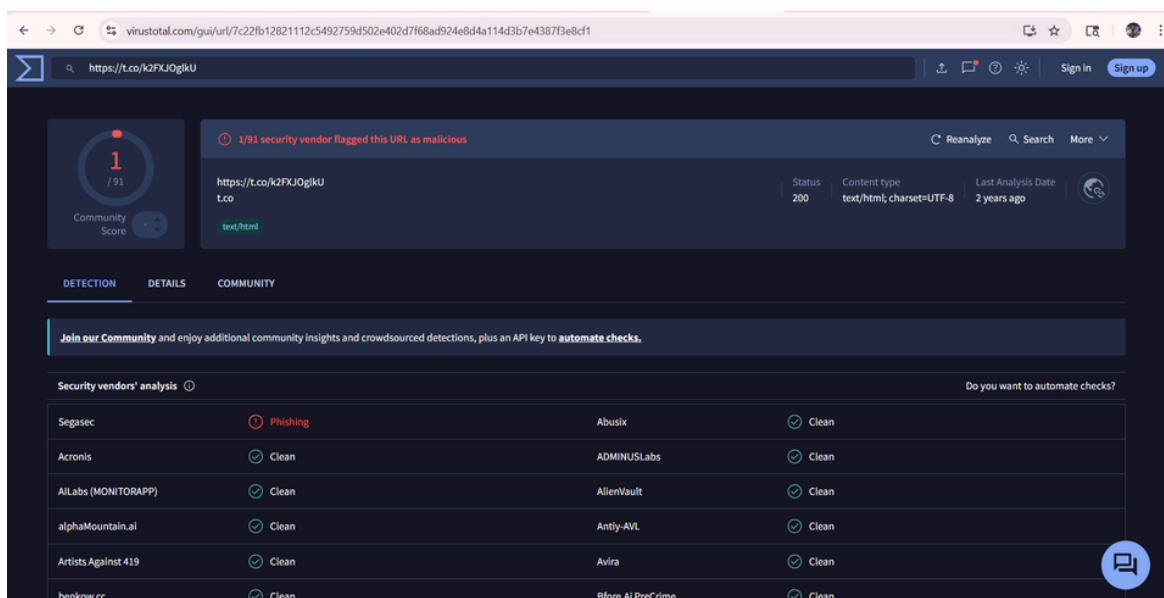


Figure 12: VirusTotal scan of unsubscribe link dtherhproblem.us — flagged Phishing/Suspicious

Indicators Found:

- Fake sender domain — ri69aezzhy.com not kohls.com
- Three different fake domains used
- SPF None — domain has zero email security setup
- DKIM None — no digital signature at all
- DMARC None — no domain record found at all
- Too-good-to-be-true prize scam — fake Ninja Foodi Grill
- Urgency language — "Act Now!", "Expiring Soon"
- Hidden redirect links — t.co shortener hides real destination
- Malicious unsubscribe link — dtherhproblem.us flagged by Fortinet
- Duplicate Content-Length — technical spam filter evasion
- Hundreds of hidden random codes — advanced evasion technique

Risk Classification: PHISHING

4. Risk Classification Summary

Email Type	Claimed Sender	Real Server	SPF	DKIM	DMARC	VirusTotal	Risk
Inheritance Scam	icloud.com	decipherhealth.in	Softfail	None	Fail	N/A	PHISHING
Payment Fraud	mercado pago.com.br	soldrun.com	Fail	Fail	Fail	7/95 Malicious	PHISHING
Fake Prize Scam	kohls.com	gybatedtioneer.net	None	None	None	1/91 Phishing	PHISHING

5. Prevention & Awareness Guidelines

5.1 Do's — What Employees Should Do:

- Always verify the sender's email domain carefully
- Hover over links before clicking to see the real URL
- Contact IT/Security team if you receive a suspicious email
- Use Multi-Factor Authentication (MFA) on all accounts
- When in doubt, call the sender on a known phone number
- Report phishing emails to your security team immediately
- Check for spelling mistakes in domain names

5.2 Don'ts — What Employees Should Never Do:

- Never click links in unexpected or urgent emails
- Never enter passwords on sites reached via email links
- Never transfer money based only on email instructions
- Never open attachments from unknown senders
- Never share OTPs, passwords, or PINs via email
- Never trust an email just because it looks official
- Never click "Unsubscribe" in suspicious emails

5.3 What To Do If You Receive a Suspicious Email:

Step 1 — STOP. Do not click anything.

Step 2 — CHECK the sender's email domain carefully.

Step 3 — HOVER over any links to see the real destination.

Step 4 — VERIFY by calling the sender on a known number.

Step 5 — REPORT to your IT/Security team.

Step 6 — DELETE the email from inbox and trash.

6. Tools Used

Tool	Purpose	Link
GitHub — phishing_pot	Phishing email samples	github.com/rf-peixoto/phishing_pot
Google Header Analyzer	Email header analysis	toolbox.googleapps.com/apps/messageheader
MXToolbox	SPF/DKIM/DMARC verification	mxtoolbox.com/EmailHeaders.aspx
VirusTotal	Malicious URL scanning	virustotal.com
Canva	Report documentation	https://www.canva.in/

7. Conclusion

This analysis of three real phishing email samples demonstrates that phishing attacks rely heavily on social engineering, domain spoofing, and authentication failures. All three emails failed SPF, DKIM, and DMARC checks — proving they were sent from unauthorized servers while impersonating trusted organizations.

The most effective defense against phishing is a combination of technical controls (email authentication, spam filters, MFA) and employee awareness. Users who know how to identify phishing indicators — fake domains, urgency language, suspicious links — are significantly harder to deceive.

Organizations that invest in regular phishing awareness training can dramatically reduce the risk of successful attacks and the financial and reputational damage they cause.